INNOTEC
high engineering

Operating Instructions

Zero-air Generator AirKat 6.1

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1 General Information

Instruments of the „AirKat“-series produce highly cleaned zero- or reference air. They are high-grade carefully conceived and constructed robust auxiliary instruments (independent modules) for the gas-measuring and analytical technique for permanent operation use in industry or laboratory applications and are process-suitable.

An important reason for purchasing is a clear reduction of operating costs by substituting gas cylinders.

Before commencement of operation this manual must be read carefully. It contains instructions and technical specifications whose observance is important for a trouble-free operation (e.g. "dry air", …, but why? …).

1.1 Warranty and Liability

A warranty period of 24 months is given starting with the delivery date. Within this period INNOTEC cures free of charge all deficiencies which are caused by material or manufacturing defects, but no deficiencies which are caused by carelessness in handling. INNOTEC will at its own discretion repair or replace the defective items or the whole unit.

Liability of any kind for damage or consequential damage, not concerning the zero-air generator, resulting from improper operation, wear or wastage (e.g. IR-heat-radiator, cooling fan, material for filter cartridge) or improper engagement is ruled out.

Any kind of manipulation of the unit without agreement of the manufacturer – with exception of proficient maintenance in accordance with chapter 7, p. 9 – has the consequence of loss of any kind of guarantee claims!

1.2 Safety Instructions

The zero-air generator must only be used in places where no danger can result from the unit. Especially an overheating must be excluded.

Warning:
The zero-air generator has *no Ex-approval* and must not be operated in explosion endangered areas. A connection with gases which might ignite or burn is strictly forbidden (danger of life).

The zero-air generator must not be fed with aggressive air which affects the gas tubing, poisons the catalyst or pollutes by condensation or dust.

It must not be tried to repair the unit or making any changes in the set-up with the exception of consultation with the manufacturer and only by trained technical personnel familiar with potential dangers and the valid regulations.

On principle attention must be paid to the following points:

- The zero-air generator must always be separated from mains on any connecting- repair- or maintenance work
- The zero-air generator must not be operated without protector ground (power cable)

- The protector ground connections inside and outside the unit must not be disconnected
- In need of repair or maintenance the service and maintenance instructions must be followed
- Only original spare-parts must be used (otherwise no claim for damages)
- Doing electrical installations (230 V) the respective national guidance is to be respected (in Germany: VDE0100)

If it is assumed that a regulated and safe operation of the zero-air generator is no longer given the zero-air generator is to take out of operation and must be blocked for unauthorized starting-up

2 Fields of Utilization

Zero-air generators generally can be employed with advantage if synthetic air is needed for the operation of gas measuring instruments or equipment as for example:

- on process applications in the industry (e.g. chemical production)
- in analytical measuring techniques as calibration and feeding air
  – for analytical gas analyzers
  – for gas measuring facilities
  – for vehicles equipped for measuring immission gases
  – for immission gases analyzing stations
- in laboratories as feeding gas for FIDs (flame ionization detectors) or gas chromatographs
- In areas where no pressure cylinders must be used

Zero-gas generators substitute cost- and labor-saving synthetic air from gas cylinders. They enable a constant very good quality of reference air and amortize themselves in numerous uses in a short period of time.

3 Operating Principle

Zero-air generators produce from **dry (!)** surrounding air which is offered by an oil-free compressor and dryer (e.g. adsorption dryer) or by a central air supply a constant qualitatively high-grade zero-air resp. reference air from which all hydrocarbons, incl. methane, are removed.

Functional principle is the catalytic oxidation of hydrocarbons and carbon monoxide at high temperature to carbon dioxide and water. As catalytic material a special mixture is used which is heated in a reactor up to 430 °C with help of an IR-heat-radiator. The heat transfer via infrared-radiation makes a quick response possible and an effective heat transfer, especially at gas-flow changes. The oxidized gas is cooled down in a heat exchanger or a stainless steel helix with blower and is led through a filter cartridge (standard filling: molecular sieve 10 Å) whereby also water vapor and carbon dioxide is removed out of the gas stream (but only with correct handling, see catchword “dry air”).

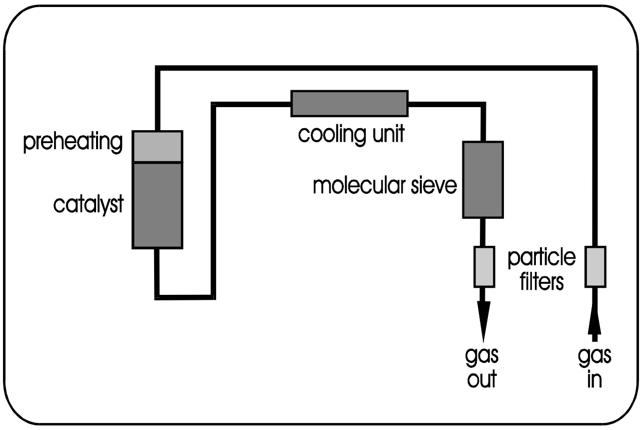


Fig. 1 Flow chart

4 Heat reactor

The reactor as „heart“ of the AirKat has stainless steel material (V4A) and was conceived and constructed very carefully, so that it quickly responds to flow changes and the operating conditions are always met (also at extreme flow changes, e.g. from 0 to 6 l/min). The fed gas flow is heated before reaching the catalyst material so that even at full flow no cooling effect occurs and a full efficacy is ensured. The catalyst material (standard: mixture of Palladium and Platinum) is heated uniformly by a heating radiator (320 W) and is distributed homogeneously without formation of channels. ***The heating radiator is accessible from the housing front****! (In case of changing the radiator please contact us!)*

The reactor (no replacement part) is the part of greatest expenditure of the zero-air generator. Therefore care must be taken that no gases are applied that act as catalyst poison and reduce the effectiveness.

Using the AirKat appropriately the catalyst-material does not alter and remains effective over many years. Therefore the AirKat is a source of permanent constantly high quality zero-air (resp. reference-air) which as substitute of gas-cylinders can amortize itself shortly.

5 Unit Set-up

5.1 Front View

The AirKat is switched-on and off with help of a push-button (red=operation) on the front plate.

A three-digit digital display shows the actual temperature of the reactor.

The status of the unit is perceived at a glance at the two LED‘s „OPERATION“ and „FAULT“ (see chapter 6.4 „Readiness for Operation“, p. 8).

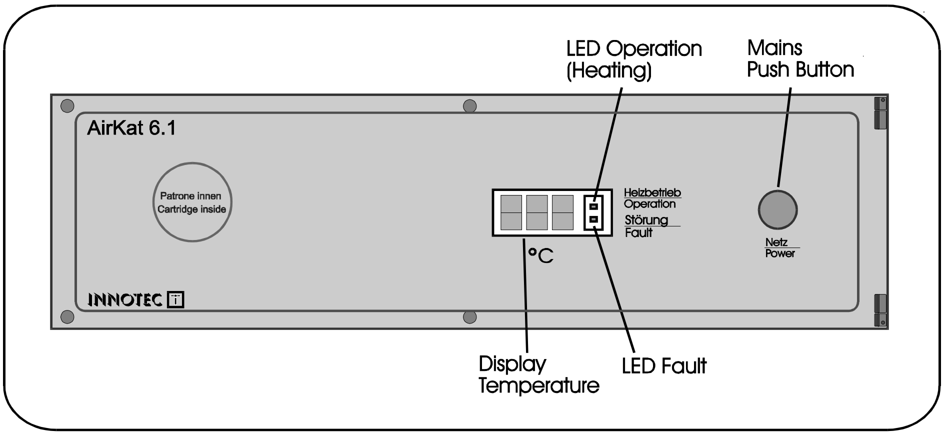


Fig. 2 Front view

5.2 Filter Cartridge

The high-grade filter cartridge (V4A) is accessible after opening of the swivel front-plate. It is refillable and has a volume of 640 ml.

By choosing special fillings, e.g. molecular sieve or particle filter, also special demands for preparing gases can be taken into account without additional modules or devices.

The standard filter material are molecular sieve pellets ((pore size 10 Å) for removing the reaction products CO_2 and H_2O.

Attention:

The AirKat must be operated only with *dry and clean air* because otherwise the filter cartridge (also for CO_2) quickly gets inactive.
A connection with any other gases like pneumatic air for cleaning (humidity!), e.g. for quicker cooling down of the reactor or in times, when the catalyst is not heated up, is not allowed.

5.3 Rear View

The gas connections for in- and output are found on the unit rear plate at the two particle filter cylinders (pore size 5 µm; 6 mm compression fittings), also the alarm-contacts, the mains connection with electronic filter and fuses and a blower-output for cooling the unit.

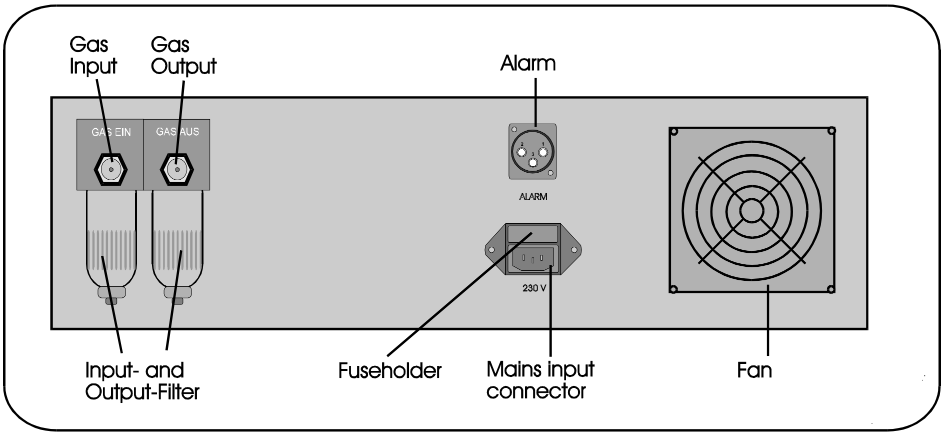


Fig. 3 Rear view

5.4 Overheat Protection

The zero-air generator is equipped with a temperature protection fuse.

The temperature fuse (red; not electronically) at the electronics-lining beside the reactor housing serves as overtemperature protection of the unit-inside. If the temperature inside the unit rises over 77 °C, the temperature protection responds and interrupts the power to the heating radiator. The temperature indication and the light diode OPERATION are still active, that means normal operation is indicated. The reactor cools down under the lower temperature limit (as seen in the temperature reading) and the alarm relay switches. The temperature fuse must be replaced.

Attention:

If the temperature fuse responds the cause of the failure of overheating must be cleared up before the fuse is replaced and the Air Kat is restarted. For changing the fuse the cover plate must be removed.

—> in any case the power must be cut!

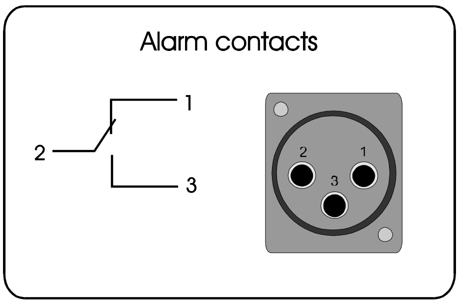
5.5 Heat Regulator and Alarm Contacts

The zero-air generator is equipped with a **microprocessor controlled PID regulator**. A display on the front-plate (Fig. 2, p. 5) shows the reactor temperature and the actual status of the regulator.

The switching of the heat-radiator is effected from a semiconductor relay, resulting in a long lifespan of the temperature controlling circuit.

To protect the electronics from voltage peaks and irregularities of the power supply system the unit mains-input is equipped with an electronic filter.

A **threshold-regulation** controls the under- and overtemperature of the reactor (nominal temperature: 430 °C) and switches relay-contacts at the **alarm-socket** (Fig. 3, p. 5) if the temperature is lower or higher than the alarm points (380 °C resp. 530 °C). These potential-free relays function as opening and closing contacts (position 1, 2 and 3 are contacted; s. Fig. 4). Within the nominal temperature interval (380–530 °C) the contacts position 1 and 2 are open (they close on alarm) and the contacts position 2 and 3 are closed (they open on alarm). If the actual temperature lies outside the nominal temperature interval (e.g. during the heating-up phase) the contacts position 1 and 2 are closed and the contacts position 2 and 3 are open. The contacts are designed for Ohm-loads.



max. constant current:		2 A
max. switching voltage:	AC voltage	250 V~
	DC voltage	220 V–
max. switching capacity:	AC voltage	62,5 VA
	DC voltage	30 W

Fig. 4 Connection of alarm contacts

Attention:
If the unit is operated without surveillance provisions must be made for guaranteeing a control.

6 Starting-up the Zero-air Generator

Attention:
Starting-up the unit must only be done under observance of this operating instructions and only by qualified personnel.

Any compressed-air must be dry (!), dust- and oil-free.

A connection with any other gases like pneumatic air for cleaning (humidity!), e.g. for quicker cooling down of the reactor or in times, when the catalyst is not heated up, is not allowed.

6.1 Installation

In installing or setting up the unit care must be taken that the outgoing warm air from the blower (rear front, Fig. 3, p. 5) is not hindered.

The unit must be protected against water splashes and solar radiation. The allowable surrounding conditions (15 °C to 35 °C) must be observed.

A certain transmission of heat from the unit has to be taken into consideration so that an adequate distance to other instruments or installations must be kept.

The base-knobs can be easily unscrewed on the outside as required.

6.2 Gas Connections

Attention:
The AirKat must not be operated in explosion dangered areas.

A connection with gases which might ignite or burn is not allowed.
Gas-input and gas-output must not be changed by mistake.
No fluids must get into the catalyst.

Any compressed-air must be dry (!), dust- and oil-free.

The maximum allowed operating pressure and the maximum gas-flow must not be exceeded (see technical data).

- Connect the gas-input (Fig. 3, p. 5) with the central compressed-air supply resp. with an oil-free compressor with air-drying module.
- Check for a correct connection and test for leaks.
- Connect the gas-output not before a **running in time** (see below) with the corresponding fittings of the final instruments.

6.3 Electrical Connections

Attention:
This unit works with voltages up to 240 V above the ground potential. Therefore special care must be taken for a contact protection.

In electrical installing works the valid national regulations must be observed (in Germany VDE 0100).

Connect the power cable with the mains-socket of the zero-air generator (Fig. 3, p. 5) and a wall socket with ground contact.

The ground-protection must not be removed by cables without safety contacts.

6.4 Readiness for Operation

Attention:
Make sure that all instructions are observed before starting the unit and that the values of the technical data are followed.

The AirKat has been **preconditioned** in factory. But for reaching the full functional capacity a **running-in time** should be regarded depending on the application conditions and the (pre)-stress, that means operation with dry air without connection to the final instruments.

The catalyst material of the reactor (and the filling of the filter cartridge) are dry at delivery. If the unit is fed with humid air and in the worst case only later the end-temperature of the reactor is reached, condensed water can develop in the cooling helix and even collect! The filling of the filter cartridge then is wet and therefore inefficient. Such a case is grossly negligent and falls not within the guarantee!

- Switch on the unit by pushing the mains button (Fig. 2, p. 5). Both LEDs (LED „OPERATION“ and LED „FAULT“, Fig. 2) shine red. The reactor is heated up in about 30 minutes.

  Take notice of:

  - The **LED OPERATION** is on if the heat radiator is in operation during a heating interval (for different long times according to the actual temperature); it is off for the rest of the interval.

  - The **LED FAULT** is on if the alarm points (380 °C and 530 °C) are under- or overstepped; the unit is not ready for operation at a temperature less than 380 °C. The LED goes out if the reactor temperature lies between the alarm points; the unit now is ready for operation, the alarm (alarm socket, s. Fig. 3, p. 5) goes out.

- **The temperature regulates itself at a value of about 430 °C. During the operation it can lie between the lower and upper alarm point (380 °C resp. 530 °C)!**
- Release the compressed air supply.
- After the running-in phase a qualitatively high-grade zero-air is available. Only now release the gas-output (Fig. 3).

7 Maintenance Works

> **Attention:**
> **If the alarm contacts are not used the display of the zero-air generator must be controlled in regular intervals to recognize malfunctions.**
> **In regular intervals the blower must be controlled for operation (cooling of the unit).**

Additional to these maintenance tests important for a correct function, the filters of the AirKat must be looked for.

- The **filter material of the stainless steel filter cartridge** generally should be changed every 6 months. As well the O-ring must be checked and if necessary be changed.
- The **gas input and output particle filters** (unit rear) are to be controlled in regular time intervals for pollution (visual control) and if necessary the filter elements and the O-rings must be changed. The maintenance-free time essentially depends on the quality of the fed compressed air.

7.1 Changing the Material of the Stainless Steel Filter Cartridge

> **Attention:**
> **A blower breakdown may cause a hot cartridge.**
> **Don't open the cartridge under pressure. Do not grease the screw threads and the O-ring.**

The stainless steel filte cartridge with a volume of 640 ml is refillable!

Advantage: The cartridge can be refilled with different materials adjusted to the respective demands.

The filter material (standard: molecular sieve 10 Å) can be exchanged without removing the unit; just the front-plate must be opened to one side (right hand):

- Pull the mains plug.
- Disconnect the air supply and depressurize the tubing.
- Loosen the screws at the front plate, swing the front plate to the side.
- Loosen the fixing screw for the cartridge above.
- Disconnect the Teflon tubing at the front side (well accessible).
- Take out carefully the filter cartridge to the front side by slight turning.
- Open the cover-plate of the cartridge and exchange the filter material.
- Eventually replace a new Viton-O-ring.
- Do not grease the O-ring.
- Close the filter cartridge and build it in. ***Please fasten the fixing screw lightly***!

7.2 Changing the Filter Elements of the Input/Output Particle Filters

- Disconnect the air supply and depressurize the tubing.
- Unscrew the filter cylinder.
- Unscrew the finefilter element by hand and replace it; if necessary replace new O-rings.
- Take care of the right fit of the O-rings¸ **do not grease the O-rings!**
- Screw the filter cylinder in.

8 Technical Data

Function principle :	catalytic oxidation at high temperatures
Output concentration :	< 0,1% of input concentration (max. 100 ppm methane)
Input pressure :	max. 6 bar
Gas flow :	max. 6 l/min
Temperature reactor :	430 °C (+100 °C / -50 °C)
Warm up time :	appr. 20 min
Temperature regulation :	principle: microprocessor controlled 2 point regulator with PID-feedback; adjusted special electronics by INNOTEC, with optional additional functions
Temperature indication :	3-digits for temperature, one LED for operation (on during reactor is heated), one LED for failure
Controlling :	potential free contact for controlling the operation max. 250 V (AC) / 2 A
Power supply :	220 V....240 V / 50 Hz
Power consumption :	appr. 240 W during heating-up (1 heat radiator, 320 W), appr. 110 W mean at permanent operation (max. flow-through)
Fuse :	2 fuses M 3,15 A
Filter Cartridge : (molecular sieve)	stainless steel, volume: 640 ml, standard filling material: molecular sieve pellets, 10 Å pore size
Particle filters :	2, for gas input and gas output, pore size 5µm
Compression fittings :	6 mm compression fittings, stainless steel
Protection class :	IP 20
Surrounding temperature :	15–35 °C

Enclosure :

	rack mount
height	132 mm (3 HE)
width	436 mm (19"; 83 TE)
depth	380 mm (+ 40 mm for grips)

Weight :	appr. 10 kg

9 Replacement Parts

Product-No	Product
AKE.1012	High temperature radiation element 230 V, 320 W
AKE.2020	Temperature sensor
AKE.2030	Temperature fuse
AKE.3010	Cooling fan 80/80
AKE.3020	Semiconductor relay 10 A
AKE.4010	Molecular sieve material (for example 10 Å) for re-filling of the stainless steel filter cartridge
AKE.5010	Viton-O-rings for the stainless steel filter cartridge (1 O-ring/cartridge)
AKE.4020	Filter elements for gas input and output particle filters (1 element/filter; 2 elements/AirKat)
AKE.5020	Viton-O-rings for gas input and output particle filters (1 O-ring/filter; 2 O-rings/AirKat)

INNOTEC reserves the right to make modifications of their products in keeping with further technical advances.